CULTIVATION GUIDE | WEATHER

Weather and Vegetation Change in Paddy Fields

Six patterns for reading a change in greenness against recent and forecast weather. Weather context makes a change intelligible; it does not establish its cause.

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Teaching edition for the STEMLink AI Engineer Bootcamp. Content restructured from AgriVibe knowledge base AV-KB-001 Rev 1.1, which grounds its agronomy in Department of Agriculture and RRDI guidance for Sri Lanka. Not for use with real farmers until the verification items in AV-POL-03 are closed.

1. Correlation patterns

Pattern 1: Declining NDVI with heavy recent rainfall

Possible interpretations: excess water or poor surface drainage may be affecting the crop; runoff over the field bunds may have carried away applied nitrogen; flooded sections may depress the field-level measurement even where plants are healthy; lodging may have occurred, the crop having been flattened by rain or wind. In Wet Zone and Intermediate Zone fields, prolonged flooding can also worsen iron toxicity, which presents as reddish-brown or bronzed leaves rather than the even pale yellow of nitrogen shortage. Field inspection is necessary before any cause is accepted.

Suggested wording: "The vegetation decline occurred during a period of heavy rainfall. Excess water or drainage problems may be contributing factors, but the available evidence does not confirm the cause."

Pattern 2: Declining NDVI with unusually dry conditions

Moisture stress may be contributing. Tank or channel water availability should be checked, since most Sri Lankan rice depends on rainfall or supplementary irrigation rather than a guaranteed supply. Inspect the field for dry or under-irrigated sections, particularly those furthest from the water inlet. A transplanted crop is generally more vulnerable to drought than a direct-sown crop.

Suggested wording: "The vegetation decline occurred alongside low rainfall or dry conditions. Water stress may be one possible contributor, but field inspection is required."

Pattern 3: Increasing NDVI with moderate rainfall

Vegetation may be developing or recovering. The increase may equally reflect the crop's normal growth stage, particularly during tillering and up to panicle initiation.

Pattern 4: Stable average NDVI but uneven field

Overall greenness has not changed considerably, but some sections may be developing differently from others. Inspection should focus on the weaker areas: low-lying corners, poorly drained sections, and the points where water enters or leaves the field.

Pattern 5: Declining NDVI with poor data quality

The apparent decline must not be correlated strongly with weather. Further observations or a physical inspection are needed before any conclusion. During monsoon periods this situation is common and expected.

Pattern 6: Heavy rainfall forecast before a planned application

This pattern concerns the timing of an action rather than the cause of a change. Nitrogen applied shortly before heavy rain may be lost through runoff over the field bunds, particularly where surface drainage is poor. The practical response is to consider delaying the application until rainfall has passed and field conditions have stabilised, and to check drainage in the meantime. Advice on timing is within scope. The quantity to apply is not; it comes only from the Grama Niladhari division recommendation or the local Agriculture Instructor.

2. Seasonal context

Sri Lankan paddy is grown in two seasons tied to the monsoons: Maha, the main season, running roughly September to March under the North-East monsoon; and Yala, running roughly May to August under the South-West monsoon. Heavy rainfall during these periods is normal rather than exceptional, so the mere presence of rain is weak evidence. What matters is whether rainfall was heavy enough to cause flooding, runoff or drainage failure in that particular field, which only inspection can establish.

LANGUAGE RULE (ABSOLUTE)

Weather correlation generates possible explanations, never diagnoses. Permitted phrasing includes "may be related", "may be contributing", "possible explanation" and "requires field inspection". Prohibited phrasing includes "definitely caused by", "proves that" and "confirms the disease".